

Objective

A recent Electronic Engineering HND graduate seeking to build a career in electronic engineering, with a focus on circuit design and embedded systems. Combined formal study with practical project work. Work includes CMOS circuit simulation and physical IC layout, to build technical expertise. Aiming to enroll in continuous learning and applying strong problem-solving skills to real engineering challenges.

Education

Higher National Diploma, Electronic Engineering | [University of Southampton]

Dissertation Topic: DRAM Macro Cell Design. Created a DRAM cell using Computer Aided Design. Skills required for this included researching previous designs and then being able to improve upon these previous designs.

4 A-Levels | Grades (AAAB)

Maths, Physics, Economics, Chemistry

11 GCSEs + 2 AS Levels | Grades (6 A*s, 5 As, 1 B)

Further Maths, Maths, Religious Studies, Physics, Business Studies, French, Biology, Chemistry, English Literature, English Language, History, Extended Project Qualification (AS), IT (AS)

Work Experience

Celonis, Data-Driven Management (Online) (2023)

- Completed a process mining course, applying the technique to a case study business to identify supply chain efficiencies
- Achieved a 10% increase in ingredient supply for a small pizza shop case study through supply chain analysis

Private Tutor (Self-employed) (2024 to Present)

- Delivered one-to-one tuition, tailoring explanations and pacing to individual student needs
- Developed strong communication skills in breaking down technical and academic concepts clearly

Freelance AI Model Training and Evaluation (Prolific, Mercor, Amberjack) (2024 to Present)

- Contributed to AI model training and evaluation through structured task-based freelance work, including data annotation, response evaluation, and quality assessment
- Maintained consistent, self-directed output across multiple platforms while managing workload independently

Personal Projects

CMOS DRAM Cell Design and Embedded Systems Development (Self-directed) Further information on Github Website;

<https://fisla-projects.github.io>

- Extended undergraduate dissertation work on DRAM macro cell design through independent study: designed and simulated a 1T1C DRAM memory cell in LTspice, covering write and read operations, charge leakage, and refresh timing
- Diagnosed and resolved transistor-level circuit issues including threshold-voltage limitations and a parasitic body-diode leakage path, verifying results against analytical calculations
- Produced a physical transistor layout using KLayout and the open-source SkyWater sky130 process design kit, translating circuit-level design into fabricable device geometry
- Designing and building an embedded environmental monitoring system (Raspberry Pi Pico, multi-sensor integration, live display, data logging)

Skills

- Circuit simulation and CAD: LTspice (SPICE simulation), KLayout (IC layout), MATLAB, L-Edit, Simul8
- Analysis skills using charts and graphs
- IT skills: proficient in Microsoft Excel, Word, PowerPoint, Access
- Strong maths and communication skills
- Full UK Driver's License holder